

OPERATING MANUAL

AUTOMATIC POWER OFF FUNCTION
AUTOMATIC POWER ON FUNCTION
AUTOMATIC WARMING-UP FUNCTION

Manual No.: HGENSB0155E

NC: MAZATROL SmoothAi
MAZATROL SmoothX
MAZATROL SmoothG
MAZATROL SmoothG³

Serial No.:

Before using this machine and equipment, fully understand the contents of this manual to ensure proper operation. Should any questions arise, please ask the nearest Technical Center or Technology Center.

IMPORTANT NOTICE

1. Be sure to observe the safety precautions described in this manual and the contents of the safety plates on the machine and equipment. Failure may cause serious personal injury or material damage. Please replace any missing safety plates as soon as possible.
2. No modifications are to be performed that will affect operation safety.
3. For the purpose of explaining the operation of the machine and equipment, some illustrations may not include safety features such as covers, doors, etc. Before operation, make sure all such items are in place.
4. If there is a difference between the contents of the manual and the actual state of the machine, please contact the nearest Technical Center or Technology Center.
5. Always keep this manual near the machinery for immediate use.
6. If a new manual is required, please order from the nearest Technical Center or Technology Center with the manual No. or the machine name, serial No. and manual name.
7. The export of machining programs may be restricted to prevent them from being used for military purposes. The machining programs you prepared are your property, so you should manage them yourself in accordance with the law.
8. Unauthorized copying or reproduction of this manual is prohibited.

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1 INTRODUCTION

Automatic Power ON/OFF & Warming-Up is an optional function for automatically warming up the machine before use and turning it off after use.

Power saving and improvement in productivity can be expected by the use of this function.

This function consists mainly of the following three functions, which can be used in combination.

For details of each function, see the following chapters.

- Automatic power OFF function
- Automatic power ON function
- Warming-up function

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1 INTRODUCTION



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2 SAFETY PRECAUTIONS

This manual is prepared only for the function or the device which is specified in the title of this manual. Therefore, you are required to carefully read and understand the operating manual of the machine, especially the section describing safety precautions, before using or operating the function or the device.

2-1 Rules

The meanings of our safety precautions to DANGER, WARNING, and CAUTION are as follows:



DANGER

: Failure to follow these instructions could result in loss of life.



WARNING

: Failure to observe these instructions could result in serious harm to a human life or body.



CAUTION

: Failure to observe these instructions could result in minor injuries or serious machine damage.

2-2 Safety Precautions Relating to the Function/Device Described in This Manual



WARNING

- When shutting off the power manually, make sure that the machine is in the initial conditions. When the machine is not returned to the initial condition on turning OFF the power, in such case as emergency stop during operation of the ATC arm, etc., automatic power ON may result in accidents.

Note 1: When the automatic control power OFF function is used, the machine must be initialized at the end of the program before power cutoff.

Note 2: When the automatic power ON function is used, be sure to check the level of the oil such as spindle lubricating oil, oil in the hydraulic unit, etc.

Note 3: To use these functions, the main power breaker must always be kept energized. Therefore, the power line to the machine including a shop branch must be designed taking this into consideration.

Note 4: Note that some electrical power in the machine is energized during the standby condition waiting for the power ON according to the timer setting, due to the control of the system.

2

SAFETY PRECAUTIONS



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3 SETTING APPLICATION

MAZATROL SmoothAi/X/G/G³ is designed to use an application on the NC screen to make settings for automatic power ON and OFF. This application is referred to as APT (Automatic Power Timer).

This chapter describes how to use APT.

3-1 Start and Stop

APT is started with the Home screen of the NC unit. After completing all settings, close the APT window with the Close button.

Remark : There is no need to always keep APT active.

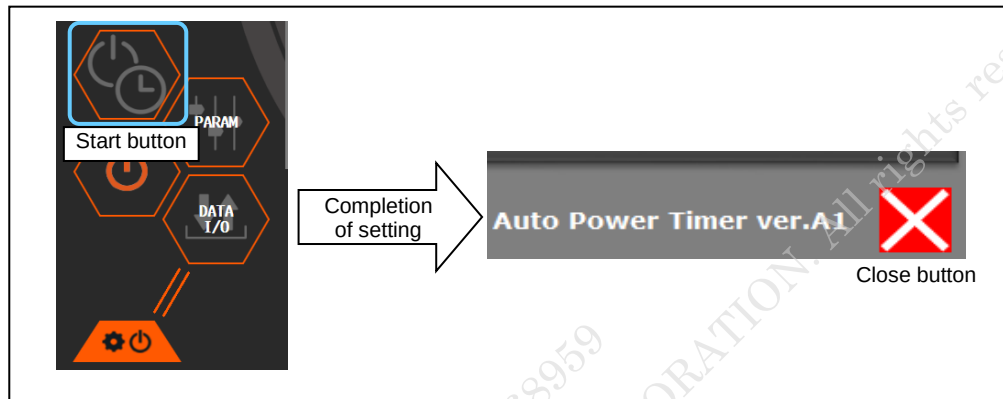


Fig. 3-1 Start button/Close button

3-2 Configuration of Displays

The APT window is displayed at the center of the NC screen.

The APT window contains two monitors: control monitor and timer monitor, which can be switched with a tab.

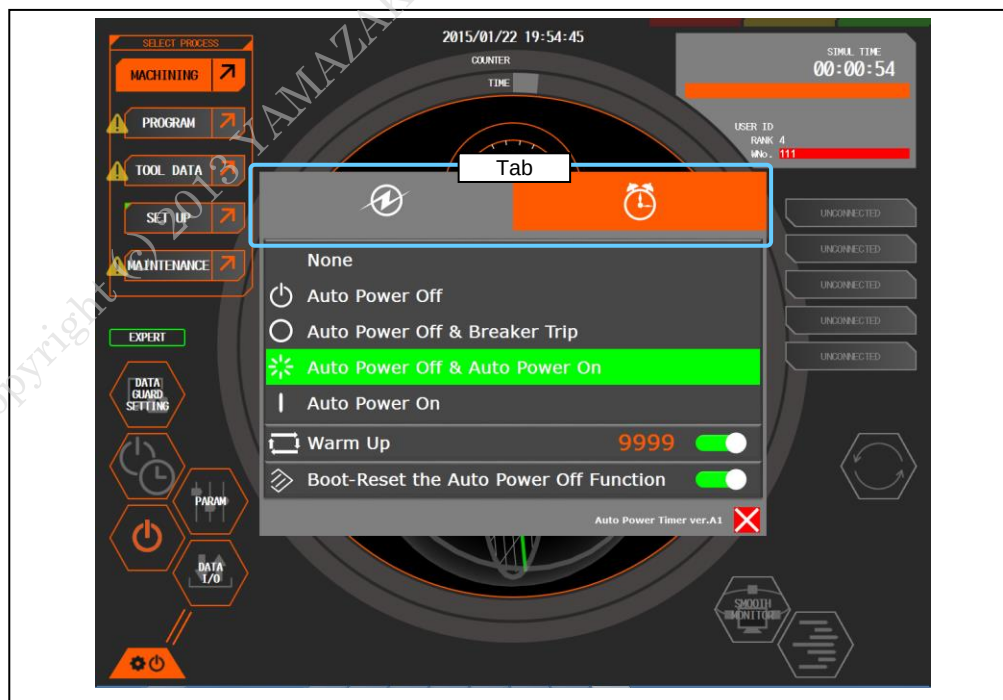


Fig. 3-2 Configuration of displays

3-2-1 Control monitor

This monitor is used to make and confirm settings for automatic power ON/OFF and a warming-up.

When the  button is tapped, this monitor is displayed.

1. Display data and functions

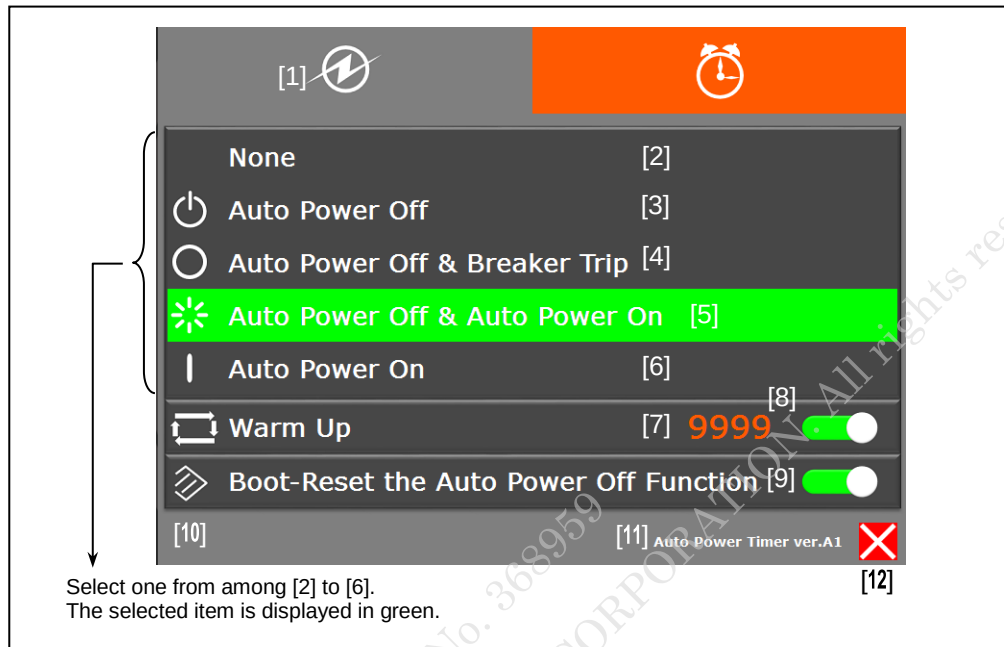


Fig. 3-3 Control monitor

Table 3-1 Contents of control monitor

No.	Name	Function
[1]	Control tab	Displays the control monitor.
[2]	None button	Deactivates the automatic power ON/OFF functions.
[3]	Auto Power Off button	Executes automatic power OFF.
[4]	Auto Power Off & Breaker Trip button	Automatically turns off the power and trips the main power breaker. Select this if you want to start the NC unit without using an external device, such as an FMS or a calendar timer.
[5]	Auto Power Off & Auto Power On button	Executes automatic power OFF and ON.
[6]	Auto Power On button	Executes automatic power ON.
[7]	Warm Up program No.	Used to select a warming-up program. Setting range: 1 to 9999
[8]	Warm Up switch	Used to enable or disable the warming-up operation after power ON.
[9]	Boot-Reset switch	Used to enable or disable the reset of the automatic power OFF function after power ON.
[10]	Alarm bar	Displays an alarm.
[11]	Version information	Displays the version of APT.
[12]	Close button	Terminates APT.

2. Data setting

Example : To make settings for automatic power OFF and ON

< Procedure >

- (1) Tap the  button.
- (2) Tap  to start APT.
- (3) Tap .
- (4) Tap .

Note : Even if the automatic power ON function is enabled, it will not be executed if the timer is disabled.



- For an explanation of how to enable the timer, refer to Subsection 3-2-2 “Timer monitor”.

- (5) To perform a warming-up operation after power ON
 - Enter a warming-up program number and turn on the Warm Up switch.



- When entering a warming-up program number, make sure that the number is within the specified range and the program is stored in the NC unit. Otherwise, no program will be called at the time of execution of a warming-up command and an alarm will be triggered.

- (6) To reset the automatic power OFF function after power ON
 - Enable the boot reset switch.



- (7) Tap  to terminate APT.

3-2-2 Timer monitor

This monitor is used to set the time at which automatic power ON is executed.

1. Display data and functions

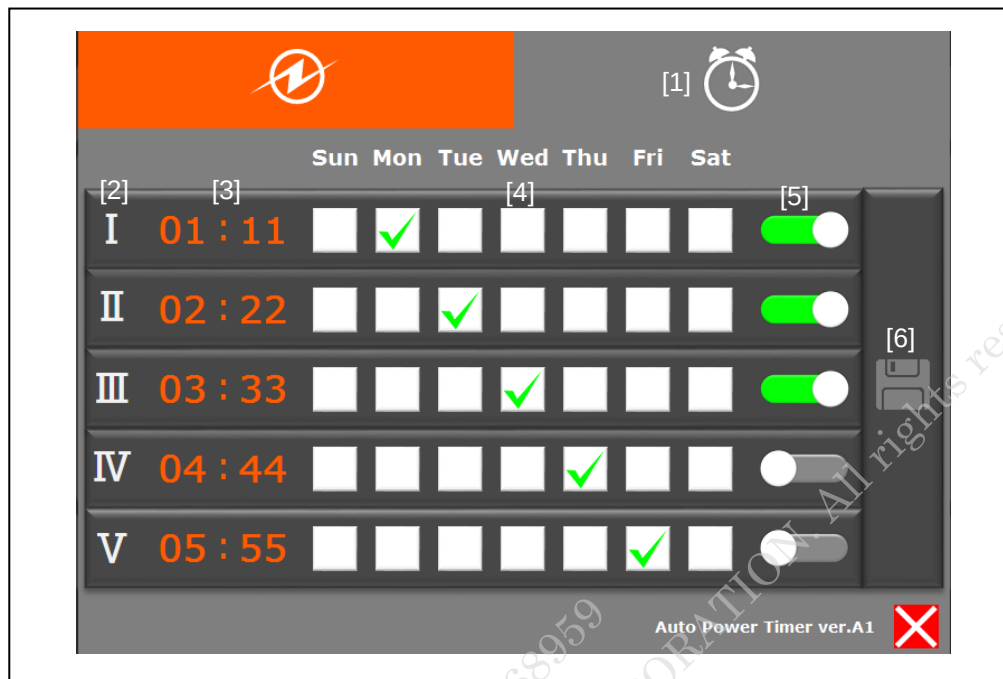


Fig. 3-4 Timer monitor

Table 3-2 Contents of timer monitor

No.	Name	Function
[1]	Timer tab	Displays the timer monitor.
[2]	Timer list	Five timer settings (channels I to V) are available.
[3]	Time	Set the time at which automatic power ON is executed.
[4]	Day of the week	Set the day of the week on which automatic power ON is executed.
[5]	Timer switch	Used to enable or disable the timer.
[6]	Save button	Used to save the timer settings.

2. Data setting

Example : To assign to Channel I, a timer that is set so as to turn on the power at 07:00 from Monday to Friday

< Procedure >

(1) Tap the  button.

(2) Tap  to start APT.

(3) Tap .

(4) Enter "07:00" in the time field of Channel I.



(5) Check the "Mon" to "Fri" boxes of Channel I.



(6) Enable the timer switch of Channel I.



Note : Even if the timer is enabled, the automatic power ON function will not be executed unless it is enabled.



- For an explanation of how to enable the automatic power ON function, refer to Subsection 3-2-1 "Control monitor".

(7) Tap .



- After setting a timer, be sure to tap  to enable the timer settings.

(8) Tap  to terminate APT.



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4 AUTOMATIC POWER OFF/ON FUNCTION AND BREAKER TRIP FUNCTION

4-1 Automatic Power OFF Function

The automatic power OFF function is designed to shut down the NC unit.

This function does not trip the main power breaker, so use it if you want to keep the NC unit on standby so that it can be started next time with an FMS or any other external device.

To execute this function, select  or  in the control monitor.

1. Timing of automatic power OFF

The NC unit shuts down when the end unit in the machining program (M02 or M30 in the case of an EIA/ISO program) is executed.

2. To turn the power back ON after it was turned OFF automatically

Press the Power ON button on the NC operating panel.

4-2 Breaker Trip Function

When an external device, such as an FMS or a calendar timer, is not used, this function will not only turn OFF the NC unit, but also trip the main power breaker.

To activate this function, select  in the control monitor.

1. Timing of automatic power OFF and tripping of main power breaker

After the NC unit shuts down following the execution of the end unit in the machining program (M02 or M30 in the case of an EIA/ISO program), the main power breaker of the machine trips to shut OFF the power to the machine.

2. To turn the power back ON after the main power breaker tripped

Turn the main power breaker OFF and back ON manually, and then press the Power ON button on the NC operating panel.

4-3 Automatic Power ON Function

This function is designed to automatically start the NC unit when the time set with timer monitor is reached.

To activate this function, select  or  in the control monitor.

Note : Even if either of the above is selected, the power will not be turned ON automatically if the main power breaker is in the OFF position.

4-4 Prior Notice of Automatic Power OFF

In order to notify the operator that the automatic power off function is active, the alarm **2518 AUTO POWER OFF** is displayed before the power is automatically turned off. The power is automatically turned off in a certain time after this alarm is displayed. If the Reset key is pressed while the alarm is being displayed, the execution of the automatic power off function will be canceled.



- For details of the alarms, refer to Subsection 6-2-2 "PLC machine control alarm".



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5 WARMING-UP FUNCTION

The warming-up function is designed to automatically execute the selected warming-up program when the NC unit is turned ON.

This function is performed when the NC unit is started with the automatic power ON function. The warming-up program is not executed when the NC unit is started manually.

5-1 Warming-Up Function of a High Speed Spindle Model

In order to perform warming-up for the high speed spindle, the warming-up program is called and executed after automatic power ON, and then monitoring of the main spindle load is performed.

In that case, a preset program may be executed after power ON, instead of the warming-up program specified with APT.



- Refer to the separate Operating Manual of the machine concerned for the details.

5-2 Warming-Up Program



- Refer to the separate Programming Manual to construct a program to your specifications.

5-2-1 Program example with MAZATROL language

Program example with MAZATROL language is shown below.

(This program example is for machining centers and contains only move commands of X-, Y-, and Z-axis (4th axis), along with spindle rotation commands.)

Example :

UNo.	MAT.	INITIAL-Z	ATC MODE	MULTI MODE	MULTI FLAG	PITCH-X	PITCH-Y		
0	CBN STL	10.	0	OFF					
UNo.	UNIT	TOOL	NOM-φ	No.					
1	MANL PRO								
SNo.	G1	G2	DATA 1	DATA 2	DATA 3	DATA 4	DATA 5	DATA 6	S M/B
1									M 39
2	90	0	X -1x					500	M 03
3	94	4	X 10						
4	1		X 0	F 1000					
5	0		Y -1y					1000	M 04
6	94	4	X 10						
7	1		Y 0	F 1000					
8	0		Z -1z					1500	M 03
9	94	4	X 10						
10	1		Z 0	F 1000					
11	1		X -1x	F 1000				2000	M 04
12	94	4	X 10						
13	0		X 0						
14	1		Y -1y	F 1000				2500	M 03
15	94	4	X 10						
16	0		Y 0						
17	1		Z -1z	F 1000				3000	M 04
18	94	4	X 10						
19	0		Z 0						
20	1		X -1x	F 1000				3500	M 03
21	94	4	X 10						
22	0		X 0						
UNo.	UNIT	CONTI.	NUMBER	ATC	RETURN	WORK No.	EXECUTE		
2	END	1	0						

Note : To allow for the maximum stroke of each axis, enter data 1x, 1y and 1z in the program.
The above program is only one example. Construct a program to your specifications.
Machines equipped with SmoothG³ do not support program with MAZATROL language.

5-2-2 Program example with EIA/ISO codes

Program example with EIA/ISO codes is shown below.

(This program example is for lathes equipped with three systems and contains only move commands of X1-, Y1-, Z1-, X2-, X3-, Y3-, and Z3-axis, along with spindle rotation commands for turning spindle 1 and 2.)

Example :

G109L1	G109L2	G109L3
G90	G90	G90
G98	G98	G98
M901	M902	G0 X-1x
M3 S500.	M303 S500.	G4 X10.0
G0 X-1x	G0 X-1x	G1 X0 F1000
G4 X10.0	G4 X10.0	G4 X10.0
M4 S1000.	M304 S1000.	G0 Y-1y
G1 X0 F1000	G1 X0 F1000	G4 X10.0
G4 X10.0	G4 X10.0	G1 Y0 F1000
M3 S1500.	M303 S1500.	G4 X10.0
G0 Y-1y	G0 Y-1y	G0 Z-1z
G4 X10.0	G4 X10.0	G4 X10.0
M4 S2000.	M304 S2000.	G1 Z0 F1000
G1 Y0 F1000	G1 Y0 F1000	G4 X10.0
G4 X10.0	G4 X10.0	M950
M3 S2500.	M303 S2500.	M99
G0 Z-1z	G0 Z-1z	
G4 X10.0	G4 X10.0	
M4 S3000.	M304 S3000.	
G1 Z0 F1000	G1 Z0 F1000	
G4 X10.0	G4 X10.0	
M950	M950	
M99	M99	

Note : To allow for the maximum stroke of each axis, enter data 1x, 1y and 1z in the program.
The above program is only one example. Construct a program to your specifications.

5-3 Cares for Warming-Up



- Check the following items beforehand to perform warming-up with care not to stop warming-up program halfway.

1. Main power breaker is to be ON

Main power breaker is to be ON for performing warming-up. Only when the main power breaker is energized, the machine power can be turned on automatically and warming-up can be performed.

2. Emergency stop is not to be at work

If power is turned on with emergency stop button kept pressed, an alarm is given, and thus warming-up cannot be performed because zero point return is impossible.

3. External feed hold is not to be at work

If external feed hold is at work, zero point return cannot be made after turning on power, and thus warming-up cannot be performed.

4. Stop position of each axis when turning on power

Prepare the last machining program so that the final positions of all feed axes are near to their respective zero points; otherwise zero point return is impossible at the beginning of the next machine operation and warming-up cannot be performed.

5. Check of the rest of lubricating oil

Lubricating oil shortage causes the machine to display an alarm to make a single block stop. In other words, warming-up stops halfway. Pay particular attention to the rest of lubricating oil for slideway.

6. Check of manual door

Close the door capable of manual opening/closing. If the door is open, cycle start will not work by the operator door interlock function.

7. ATC (Automatic Tool Changer) and pallet changer are to be in a normal state

If reset stop or emergency stop is kept applied during ATC or pallet change operation, an interlock works on each axis and spindle and thus warming-up cannot be performed.

For a normal machine operation, state of the ATC and the pallet changer need not be taken into consideration. (When power is turned off after machining is finished, ATC and pallet changer have turned to a normal state.)

8. Restriction on modification to warming-up program

Warming-up program is all the same as normal machining program, and thus it is also possible to add or modify programs for ATC, pallet change, etc. However, never add pallet change command. If the command should be added, the program stops at the pallet change command. One cycle function stops the program execution by the first pallet change command.



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6 MAINTENANCE

6-1 Parameters

This section deals only with parameters common to all models.



- For an explanation of parameters specific to each model, see the parameter list.



- After the following parameters are set, the NC unit of the machine needs to be restarted.

Parameters	Setting value	Contents
IOP4 bit 0	1	Places APT on standby.
RB91 bit 0	1	Enables the automatic power ON function.

6-2 Alarm

6-2-1 Structure of the Alarm List

This alarm list is written in the following format:

No.	Message	Type of error	Stopped status	Clearing procedure	Display
[1]	[2] (, ,)	[3]	[4]	[5]	
Cause	[6]				
Action	[7]				
Device No. related to alarm	[8]	PLC address	[9]		

[1] Alarm number

[2] Alarm message

[3] Type of error

Code	Type	Description
A	Operation	A wrong key has been pressed. Or the machine has been operated incorrectly.
B	Registered data	The program or tool data includes an error(s). There are data to be checked, such as on inspection items.
C	Servo	Malfunctioning of the servo control mechanism
D	Milling spindle	Malfunctioning of the milling spindle control mechanism
E	NC equipment	System (hardware/software) error
F	Machine (PLC)	Machine failure
G	External I/O unit	Malfunctioning of external I/O unit

[4] Stopped status

Code	Status
H	Emergency stop
I	Automatic operation cannot be started.
J	Single-block stop
K	Feed stop (hold)
L	Operation continued

[5] Clearing procedure

Code	Procedure
M	Power off → Eliminate cause → Power back on
N	Eliminate cause → Power off → Power back on
O	Eliminate the cause → Press the  key on the machine operating panel.
P	Press the  key on the machine operating panel.
Q	Eliminate the cause → Tap the  button on the NC screen.
S	Tap the  button on the NC screen.
T	Eliminate the cause.

[6] Cause of alarm

[7] Action to be taken to eliminate the cause

[8] Device No. related to the alarm display

[9] PLC address related to the alarm display

6-2-2 PLC machine control alarm

No.	Message	Type of error	Stopped status	Clearing procedure	Display
2518	AUTO POWER OFF (, ,)	—	—	P	White
Cause	- When the automatic power off function was active, an automatic power off start command (M-code: M02, M30 or the like) was issued. - The power will be automatically turned off in a short while.				
Action	- The power will be automatically turned off in a certain time. - To cancel the execution of automatic power off, press the Reset key.				
Device No. related to alarm	—	PLC address	—	—	—

6-2-3 APT window alarm

If an alarm is displayed at the bottom of the APT window, take necessary action, as described below.

No.	Message	Type of error	Stopped status	Clearing procedure	Display
001	Language file initialization error (, ,)	—	—	—	White
Cause	A language file is corrupted.				
Action	Reinstall APT.				
002	Register reading error (, ,)	—	—	—	White
Cause	A register failed to be read.				
Action	Restart the machine.				
003	Register writing error (, ,)	—	—	—	White
Cause	A register failed to be written.				
Action	Restart the machine.				
004	Invalid Register State (, ,)	—	—	—	White
Cause	The setting you made is invalid.				
Action	Correct the setting.				
201	Update (, ,)	—	—	—	White
Cause	The setting of the calendar timer is now being updated. (There is nothing wrong with the machine.)				
Action	If the update of the setting is not completed within 3 minutes, restart the machine.				
202	Calendar Timer Error (, ,)	—	—	—	White
Cause	There is something wrong with the calendar timer.				
Action	Restart the machine.				

No.	Message	Type of error	Stopped status	Clearing procedure	Display
203	Battery not mounted (, ,)	—	—	—	White
Cause	- No battery is installed in the calendar timer. - The battery in the calendar timer is dead.				
Action	- Install a battery in the calendar timer. - Replace the battery of the calendar timer.				

6-3 Installation, Version Upgrade and Uninstallation

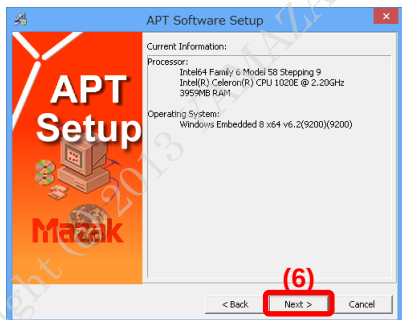
1. Installation and version upgrade

< Procedure >

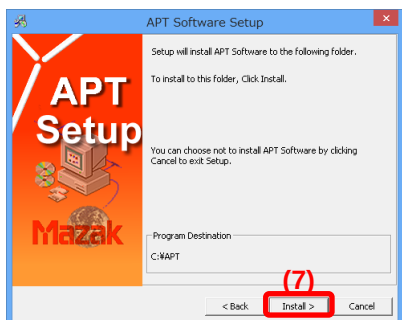
- (1) Store the installer folder "APT Installer (version name)" on a USB memory.
- (2) Exit APT.
- (3) Insert the USB memory containing the installer in the USB port of the NC unit.
- (4) Double-click "(USB memory):\APT Installer (version name)\APT Software Install.exe."
→ The installer starts.
- (5) Tap [Next].



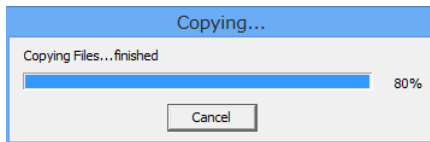
- (6) Tap [Next].



- (7) Tap [Install].



→ The installation is executed. It may take several minutes to complete the installation.



(8) Tap **[Finish]**.



(9) Create a folder under the name of “C:\APT\BACKUP.”

(10) Copy the APT Installer (version name) folder in the USB memory to the “C:\APT\BACKUP” folder in the NC unit.

Example : C:\APT\BACKUP\APT Installer A1

(11) Copy the backup data to the restoration data storage folder.



- Refer to the section “NC Backup Function” in the part 3 of the Operating Manual of the machine.

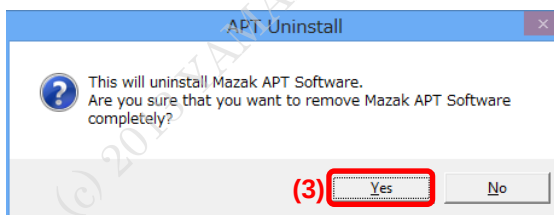
2. Uninstallation

< Procedure >

(1) Exit APT.

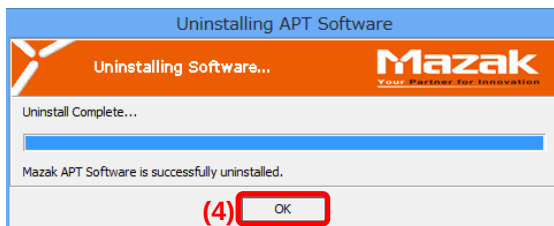
(2) Double-click “C:\APT\ APT Uninstall.exe.”

(3) Tap **[Yes]**.



→ The uninstallation is executed.

(4) Tap **[OK]**.



6-4 Replacement of Battery

The power is automatically turned ON with the clock built in the NC operating panel.

This section describes how to replace the battery of the clock when it is dead.

6-4-1 Battery life and functional restriction when the battery is exhausted

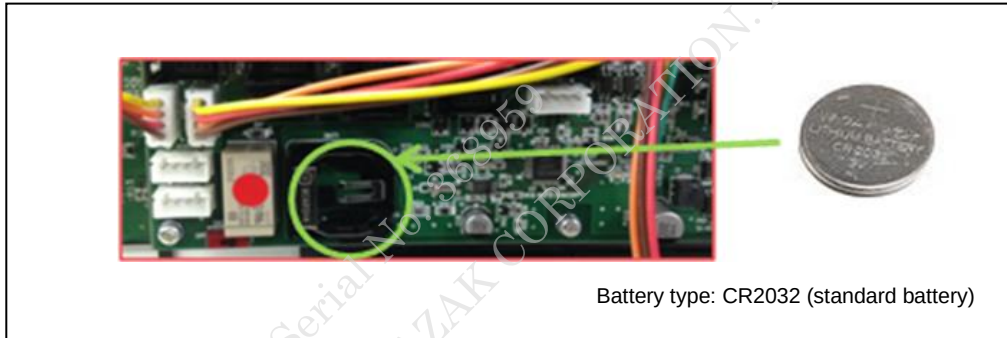
The life of the battery is about 5 years, although it depends on how the battery is used.

The battery is used to make the clock function keep working even after the main power breaker is turned OFF, so when the battery is dead, the power will not be automatically turned ON. In that case, however, the automatic power OFF function, the main power breaker tripping function and the warming-up operation function remain operative.

6-4-2 How to replace the battery

< Procedure >

- (1) Turn OFF the main power breaker.
- (2) Open the NC operating panel.
- (3) Install a battery in the battery holder at the lower part of the NC operating panel.



- (4) Close the NC operating panel.
- (5) Turn ON the main power breaker, and then press the Power ON button on the NC operating panel.
 - ➔ The built-in clock is automatically synchronized with the clock in the machine.